

Gesture2Prop: Conditional VR Prop Generation from Hand Gestures and Speech

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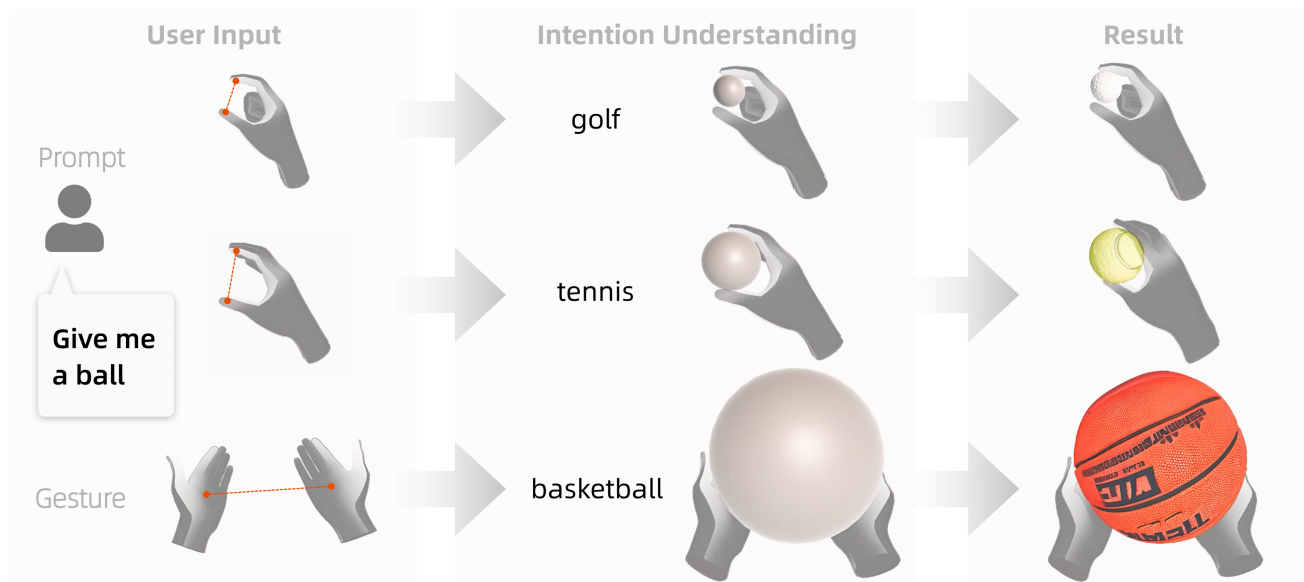


Figure 1: A method for generating VR props by estimating user intent through hand gestures and voice input.

Abstract

Virtual reality is increasingly used in education, gaming, and creative fields, yet 3D modeling remains a complex process, limiting natural and efficient interaction. While generative AI has advanced text-to-3D techniques, intuitive inputs like hand gestures are still underexplored. Inspired by everyday gesture-based expression, this paper proposes a method that combines bimanual gestures with

voice input to rapidly generate graspable 3D props in VR. By analyzing the spatial features and implicit intent of hand gestures, the system automatically infers the object's size, grip type, and interaction pose, enabling natural and well-fitted 3D model.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Gestural input*.

Keywords

Virtual Reality, 3D Content Generation, Hand Gesture Interaction

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1 Introduction

With the rapid development of virtual reality (VR) in fields such as education, gaming, and digital creation, there is a growing demand for efficient and intuitive 3D content generation tools. In immersive environments, creators increasingly seek to construct and manipulate virtual objects directly through natural body movements and spatial interactions. However, mainstream 3D modeling approaches still rely heavily on "sculpting" or "modular assembly," which are often time-consuming and lack real-time interactivity, limiting the flexibility of creative expression. Meanwhile, generative artificial intelligence (Generative AI) has made significant progress in generating images, text, and 3D models, offering low-barrier pathways for non-experts to engage in 3D creation. Yet, most existing research focuses on "text-to-model" or "sketch-to-model" techniques, with limited exploration of more natural interaction modalities in immersive settings—such as hand gestures.

This study draws inspiration from how people commonly use gestures in everyday life to intuitively convey the size and shape of objects. For example, children often mimic holding a gun or lifting a "magic orb" during pretend play. Previous research has explored various applications of gestures as interactive interfaces or retrieval methods[7, 11]. Building on these works, our study further investigates how users express creative intent and object attributes (such as size, position, and grasping style) through hand gestures.

We propose an immersive 3D prop generation method that combines hand gestures with voice input, enabling users to quickly generate virtual props in VR using simple speech and natural gestures, which are then automatically placed into their hands. Our main contributions are as follows:

- A multimodal input method combining hand gestures and voice, supported by large language models (LLMs), to efficiently infer user intent—such as prop size and interaction style—enabling a natural and rapid creation process;
- A gesture-based semi-constrained 3D model generation framework that ensures generated shapes conform to the user's hand structure while preserving form diversity, allowing for rich and expressive 3D content creation.

2 Related Work

2.1 Free-Hand Multimodal Input in Virtual Reality

Free-hand gesture interaction is a central theme in immersive user interfaces, offering intuitive and expressive ways to manipulate virtual content. Systems like VirtualGrasp[11] and Hand Interfaces[7] leverage users' real-world experience to enable natural object interactions through grasping or shape-based gestures. To further enhance immersion, recent work has explored multimodal input, particularly combining gesture with voice, which is widely regarded as a natural and direct channel for expressing user intent[1, 5]. For example, "Let's Go There"[10] shows how speech and pointing gestures can be seamlessly integrated for spatial interaction in VR. Inspired by these research works, we aim to enable users in virtual environments to express creative ideas more intuitively through the natural combination of gesture and voice.

2.2 Interaction Techniques for AI-Generated 3D Models

Current interaction methods for AI-generated 3D models mainly include text-driven[8], sketch-driven[9], and combined approaches[6]. Due to the real-time feedback capabilities of AI, users can have more creative space during the interaction process[2, 12]. For example, Yugyeong Jang et al.'s research can predict user modeling steps based on behavioral reasoning models[4]. The Style2Fab system uses AI to perform function-aware segmentation of personalized 3D models[3]. Building upon multimodal interaction research with gestures, the development of generative AI technology provides new insights for understanding voice-gesture collaborative creative intentions in VR environments.



Figure 2: Outputs under different gestures and prompts

3 Interaction And Results

To illustrate the full interaction flow (Figure 3), consider the case of a user generating a virtual firearm using both hands:

- (1) **Initial Model Generation:** When the user raises both hands in front of their chest in a gun-holding pose and says a command like "I need a gun," the system generates a basic prototype composed of primitive shapes within 2–3 seconds and automatically aligns it between the hands.
- (2) **Model Adjustment:** The user can refine the model using voice input—for example, "Make the barrel shorter, I want a lightweight submachine gun." The system responds in real time, adjusting the shape and size accordingly. Hand position and distance are also used to infer key design features, such as grip spacing.
- (3) **Final Model Replacement:** After confirming the prototype via voice, the system proceeds to generate a high-fidelity

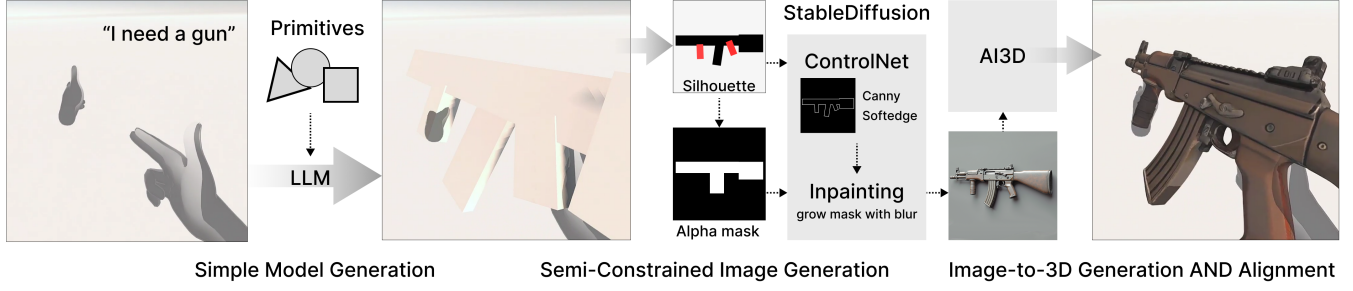


Figure 3: Our algorithmic framework takes a descriptive prompt like "gun" and dual-hand gestures as input to generate a corresponding 3D model

version with enhanced detail. Within 40–60 seconds, the refined model replaces the initial one, completing the creation process.

As shown in the figure 2, the system’s outputs under different gestures and prompts are demonstrated.

4 Implementation

4.1 Simple Model Generation Based on Gestures and Prompts

We built a VR experience and gesture recognition system on Quest 3 using Unity (version 2022.3). Leveraging the Meta OVR API, the system captures the positions of 26 hand joints in real time and compares them frame by frame with predefined gesture templates. Within a certain margin of error, it recognizes three types of hand gestures—"pinch," "grasp," and "hold"—which correspond to increasing levels of hand openness. As shown in the figure 3, we extract key distances for each gesture to estimate the initial size of the generated object.

For speech recognition, we integrated wit.ai to convert user speech into textual prompts. These prompts, together with the detected gesture data, are sent to a language model (e.g., GPT) to infer and assemble basic geometric primitives (such as spheres, cylinders, cuboids, and cones) predefined in Unity, enabling rapid construction of simplified geometric representations of the target object.

4.2 Semi-Constrained Image Generation

In Unity, we render the initial model in pure black on a white background and capture a 512×512 silhouette image to guide image generation structurally. Non-interactive regions (e.g., parts not in contact with the hand) are separated, and their bounding boxes are used to create binary inpainting masks—black backgrounds with white regions indicating areas for local regeneration.

We use ComfyUI, based on Stable Diffusion, and connect it to Unity via Python scripts. The silhouette is input into ControlNet (scribble mode) for structure control, followed by masked inpainting. Using global blending mode, the system preserves overall image consistency while allowing high variability and freedom within the masked regions.

4.3 Image-to-3D Model Generation

We used a local ComfyUI-Hunyuan3DWrapper setup, with a Unity script monitoring an input folder to trigger the pipeline in real time. The workflow includes background removal, image delighting, texture rendering, and texture sampling, ultimately producing a textured 3D model. To ensure structural accuracy, the Hy3DGenerateMesh module runs with high sampling steps and GPU memory optimization (forceoffload), while retaining the base model. Thanks to the enforced white background in earlier steps, a simplified background removal ("base" mode) suffices. To improve speed, we reduced intermediate sampling steps and limited the resolution to 512×512 during mesh generation, expanding it later for texture quality. These optimizations balance quality and performance effectively.

4.4 Model-Hand Alignment

Each prop model is preset with one or two grasp points, defining its typical holding pose and position in the local coordinate system. The primary grasp point serves as the reference for the model’s global transformation. We apply rigid alignment to match the direction vector between the two grasp points to the vector between the user’s hands. The global transformation is computed using a combination of rotation matrix and positional compensation, ensuring stable and accurate spatial alignment. For dual-hand interactions, the secondary hand’s position is dynamically adjusted using inverse kinematics (IK), allowing the model to flexibly adapt its pose based on the relative positions of both hands.

5 Conclusion and Ongoing Work

We propose a handheld prop generation method for virtual reality environments that enables the rapid creation of directly graspable 3D models based on users’ two-handed gestures and brief voice descriptions. By interpreting the implicit intent conveyed through gestures, the system extracts spatial features of the hands—such as position, orientation, and distance—as key constraints to automatically infer the prop’s size, grasping style, and interaction pose. This enables the construction of 3D forms that are both natural in shape and well-aligned with user interaction.

The core functionality of the system has been implemented, and we are currently optimizing the model generation module to better balance quality and responsiveness. To improve the generalization

of gesture understanding, we plan to conduct a pilot study to cluster commonly used user gestures and define a more representative design space. While the current system relies primarily on static gestures for intent recognition, we have observed that users often accompany imaginary object descriptions (e.g., "swinging a sword" or "drinking water") with distinctive dynamic motions that convey richer interactive cues. In future work, we aim to incorporate dynamic gesture recognition to enhance the modeling of complex interaction behaviors.

References

- [1] Jin Bai, Mohd Shahrizal Sunar, and Norhaida Mohd Suaib. 2025. Augmented reality interaction: a comprehensive review of gesture and speech integration techniques. *Neural Computing and Applications* (2025), 1–31.
- [2] Chen Chen, Cuong Nguyen, Thibault Groueix, Vladimir G Kim, and Nadir Weibel. 2024. MemoVis: a GenAI-powered tool for creating companion reference images for 3D design feedback. *ACM Transactions on Computer-Human Interaction* 31, 5 (2024), 1–41.
- [3] Faraz Faruqi, Ahmed Katary, Tarik Hasic, Amira Abdel-Rahman, Nayeemur Rahman, Leandra Tejedor, Mackenzie Leake, Megan Hofmann, and Stefanie Mueller. 2023. Style2Fab: Functionality-Aware Segmentation for Fabricating Personalized 3D Models with Generative AI. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–13.
- [4] Yugyeong Jang and Kyung Hoon Hyun. 2024. Advancing 3D CAD with Workflow Graph-Driven Bayesian Command Inferences. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–6.
- [5] Tanja Kojic, Maurizio Vergari, Maximilian Warsinke, Danish Ali, Sebastian Möller, and Jan-Niklas Voigt-Antons. 2025. Multimodal User Experience in Extended Reality: Exploring Hand Tracking, Voice, and Passthrough Interactions. In *Proceedings of the 17th International Workshop on Immersive Mixed and Virtual Environment Systems*. 8–14.
- [6] Seung Won Lee, Tae Hee Jo, Semin Jin, Jiin Choi, Kyungwon Yun, Sergio Bromberg, Seonghoon Ban, and Kyung Hoon Hyun. 2024. The Impact of Sketch-guided vs. Prompt-guided 3D Generative AIs on the Design Exploration Process. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [7] Siyou Pei, Alexander Chen, Jaewook Lee, and Yang Zhang. 2022. Hand interfaces: Using hands to imitate objects in ar/vr for expressive interactions. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–16.
- [8] Ben Poole, Ajay Jain, Jonathan T Barron, and Ben Mildenhall. 2022. Dreamfusion: Text-to-3d using 2d diffusion. *arXiv preprint arXiv:2209.14988* (2022).
- [9] Yulin Shen, Yifei Shen, Jiawen Cheng, Chutian Jiang, Mingming Fan, and Zeyu Wang. 2024. Neural canvas: Supporting scenic design prototyping by integrating 3d sketching and generative AI. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [10] Jaisie Sin and Cosmin Munteanu. 2021. Low-level Voice and Hand-Tracking Interaction Actions: Explorations with Let's Go There. In *Adjunct Publication of the 23rd International Conference on Mobile Human-Computer Interaction*. 1–3.
- [11] Yukang Yan, Chun Yu, Xiaojuan Ma, Xin Yi, Ke Sun, and Yuanchun Shi. 2018. Virtualgrasp: Leveraging experience of interacting with physical objects to facilitate digital object retrieval. In *Proceedings of the 2018 Chi conference on human factors in computing systems*. 1–13.
- [12] Chong Zeng, Yue Wu, Zisen Zhou, Yanju Yao, and Hongji Yang. 2024. Computational Imagination in Three-Dimension Scene Synthesis. In *2024 IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C)*. IEEE, 1251–1257.